VT-MUSE: Multimodal Unified Sequential Visuotactile Representation Learning for Manipulation

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***Abstract*— Contact-rich manipulation requires global visual context and local tactile feedback, yet common fusion policies mainly encode the current observations and therefore lose information when contact occludes the scene and common representation learning methods tend to only model the single modality, which lose the joint relation of vision and tactile. We present VT-MUSE, a compact two-stage framework for learning sequential visuotactile representations with multimodal unified proxy learning before policy training. First, visual and tactile encoders are adapted with cross-modal alignment, temporal contrast, and mask-consistency objectives. Second, a conditional latent model uses tactile history to recover masked recent images and tactile depth changes. Its deployable prior is integrated into an Transformer-based action generation model through gated cross-attention. On four UniVTAC tasks, VT-MUSE reaches a 62.33% average success rate, compared with 51.33% for the strongest complete baseline. On real world tasks, VT-MUSE also showcase better performance with 95% average success rate. Ablations show that temporal alignment and both reconstruction targets are important.**

I. INTRODUCTION

Vision provides object- and scene-level context for robot manipulation, while touch directly measures contact, deformation, and slip. Existing visuotactile policies have demonstrated the value of multimodal feedback [1], [2], [3], but many methods fuse independently encoded observations at the current time step and modeled the modality solely. This design can underuse the rich information between the modality and among the sequence.

VT-MUSE instead learns a temporally grounded state representation jointly from synchronized visual and tactile histories. It trains with hidden recent visual tokens. The model must therefore use contact dynamics and earlier cross-modal context to infer the current global state to inately learn the relation between two modality. We separate representation learning from policy learning so that the same pretrained encoder can support a lightweight downstream controller with comparatively few demonstrations.

Our contribution is a simple two-stage recipe. Stage I aligns pretrained visual and tactile encoders across modalities, neighboring states, and independently masked views. Stage II trains a conditional latent model to reconstruct the masked visual tail and predict tactile depth flow. At deployment, only the conditional prior is retained and fused into an ACT-style policy through gated cross-attention. Experiments on four simulated tasks and two exploratory real-world tasks indicate that this representation improves contact-rich manipulation; compact ablations isolate the benefits of the reconstruction and temporal objectives.

II. RELATED WORK

Recent tactile encoders learn transferable features through self-supervision, masked reconstruction, or robot interaction [4], [5], [6]. Cross-sensor transfer remains difficult because optical appearance varies with sensor geometry and elastomer design; transferable tactile Transformers and sensor-efficient representations address this variation directly [7], [8]. Other work combines tactile observations with neural scene representations for state estimation [9]. These methods motivate reusable touch features, but most do not explicitly ask whether a tactile sequence can recover a recently hidden visual state.

Cross-modal methods align touch with vision [10] or use attention to fuse both streams for control [2], [11]. Robot-free interfaces and policy benchmarks have further improved the scale and comparability of contact-rich learning [12], [13]. TacSL and UniVTAC provide scalable sensing and policy evaluation in simulation [14], [15]. VT-MUSE is complementary: rather than aligning only synchronized frames or fusing only current observations, it trains tactile history to compensate for masked recent vision and exports the resulting temporal state to a standard action-chunking policy.

III. METHOD

A. Problem Setup

Our data consist of trajectories $\tau = \{(o_t, a_t)\}_{t=1}^{T}$, where $o_t = (I_t, T_t, q_t, c)$ contains an external RGB image, bilateral tactile images, robot proprioception, and a discrete task identity. For each target time, the representation encoder receives a temporally subsampled sensory window and outputs f_t^{VT}. The downstream tiny policy backbone combines this feature with the current observation and predicts a length-K_A action chunk.

B. Sequential Visuotactile State

For a target time t, we sample a length-L history of external RGB images $\mathbf{I}_t$ and bilateral optical tactile images $\mathbf{T}_t$. Separate pretrained Vision Transformers encode the two modalities. Every token receives a modality embedding, a temporal-slot embedding, and a relative-time embedding. A task token conditions the shared representation when multiple manipulation tasks are used for pretraining.

As summarized in Fig. 1, Stage I adapts the final three blocks of both modality encoders. Synchronized visual and tactile tokens are positives in a symmetric InfoNCE loss $\mathscr{L}_{\mathrm{cross}}$. A second contrastive loss $\mathscr{L}_{\mathrm{temp}}$ brings neighboring multimodal states from the same trajectory together, while

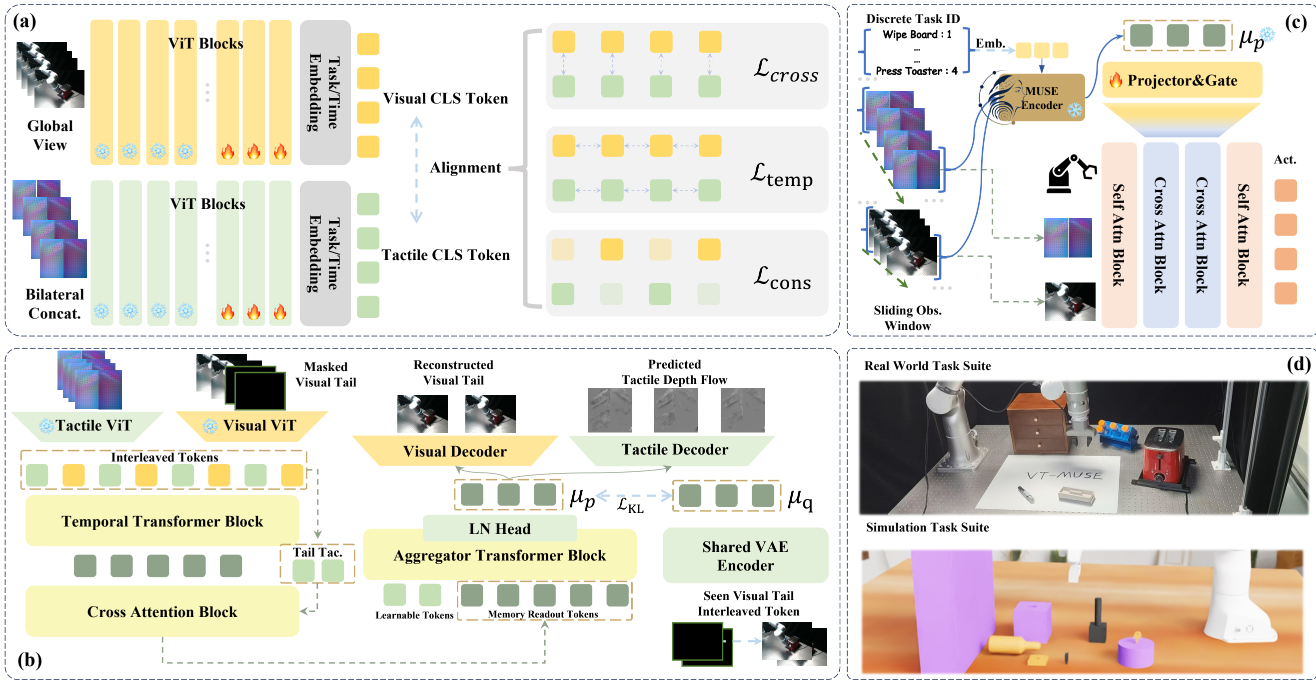

Fig. 1. **Overview of VT-MUSE**. (a) Stage I adapts separate pretrained visual and tactile ViTs through synchronous cross-modal alignment, adjacent-state temporal contrast, and consistency under independently sampled visual-token masks. (b) Stage II freezes both modality encoders and randomly masks the tail visual slots. Interleaved frame-level tokens form a task-conditioned temporal memory queried by recent tactile tokens. At this stage, we train the cVAE with tail visual reconstruction and tactile depth flow prediction losses to capture visual information and contact-rich dynamics. (c) The frozen representation is projected into memory tokens and fused into the middle ACT encoder layers through gated cross-attention.

$\mathscr{L}_{\text{cons}}$ makes two independently masked versions of one window consistent:

$$\mathscr{L}_{\text{S1}} = \lambda_c \mathscr{L}_{\text{cross}} + \lambda_t \mathscr{L}_{\text{temp}} + \lambda_m \mathscr{L}_{\text{cons}}. \tag{1}$$

This stage supplies aligned features before any action supervision is introduced. Negatives are drawn from other samples in the training batch, including samples gathered across workers. The temporal objective operates on the average of synchronized visual and tactile tokens. For mask consistency, two visual masks are sampled independently for the same sensory window. These three signals serve distinct purposes: cross-modal alignment identifies simultaneous contact, temporal contrast preserves short-term evolution, and consistency discourages the representation from depending on one particular visible patch.

C. Masked Latent Modeling

Stage II freezes both encoders and replaces the final K visual slots with learned mask tokens; the complete tactile history stays visible. Visual and tactile tokens are interleaved in temporal order and processed by a Transformer memory. The last K tactile tokens query this memory, so the latent readout emphasizes the recent contact state while retaining earlier scene context.

We use a cVAE latent model. Its prior receives only the masked visual sequence, tactile history, and task identity. During training, a privileged posterior additionally sees the ground-truth visual tokens at the masked positions. A shared RGB decoder reconstructs the missing image tail, and a tactile decoder predicts bilateral depth changes between adjacent observations. The objective is

$$\mathscr{L}_{\text{S2}} = \lambda_I \mathscr{L}_{\text{rgb}} + \lambda_D \mathscr{L}_{\text{depth}} + \beta D_{\text{KL}}(q_\phi \| p_\theta). \tag{2}$$

RGB recovery preserves global geometry, whereas depth-flow prediction emphasizes local contact dynamics. At inference, the posterior and both decoders are discarded; the prior mean is used as a deterministic state representation. The asymmetric prior–posterior design is important for deployment. The posterior may use the hidden target images to shape the latent distribution during training, but the prior is explicitly optimized to infer the same state from information that is available online.

D. Policy Integration

The frozen VT-MUSE representation is projected through a two-layer adapter and provided as a separate memory to an lightweight Transformer policy. For action tokens X and projected memory U, fusion is

$$X' = X + g\,\text{CrossAttn}(X, U, U), \tag{3}$$

where g is a learned scalar initialized to give the auxiliary memory a small initial influence. The policy predicts an action chunk and is trained with the standard action reconstruction and latent regularization losses [16]. The policy trunk takes in current observation and robot state for instant perception. We freeze the representation encoder throughout policy training. And the policy is only trained with one task each time.

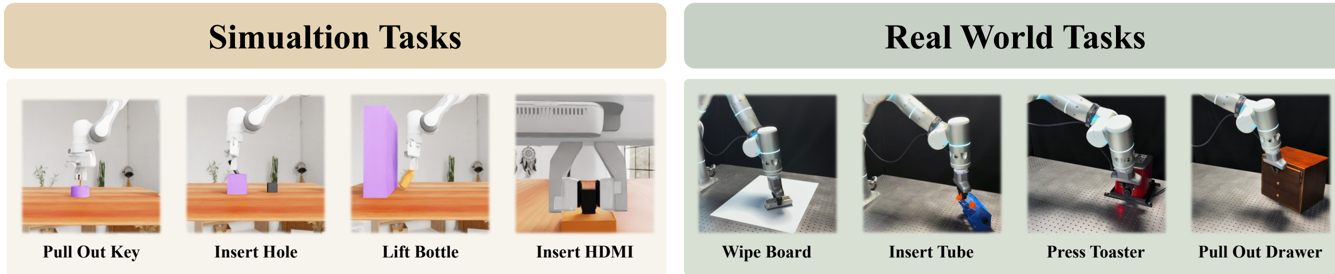

Fig. 2. We test our method both on UniVTAC simulation platform and real world setting, in which the simulation suite uses Franka Panda and Gelsight Mini and the real world benchmark use Flexiv Rizon4 along with XenseGripper.

IV. Experiments

A. Setup

We use four contact-rich tasks from UniVTAC [15] and four in real world setting as shown in Fig. 2. Together, these tasks cover grasp stability, constrained motion, contact-based geometry estimation, and precision insertion.

Each simulation task provides 500 pretraining episodes while real world task provide 50 for pretraining; policies use 50 demonstrations per task and are evaluated on the same 100 seeds in simulation and 20 simular trails in real world. Representation pretraining takes approximately 25 hours on eight NVIDIA A800 GPUs, while one policy trains in about 30 minutes on a single A800. We compare against vision-only ACT, the official ACT+UniVTAC fusion baseline, ViTaL, and FTP-$\pi_{0.5}$ without large-scale pretraining [17] in UniVTAC benchmark. The models are deployed on an NVIDIA RTX 4090. We set up comparison with ACT and ACT with tactile observation in real world setting. The models are deployed on an NVIDIA RTX 4060, and we report rollout success rate (SR).

B. Manipulation Results

Table I shows that VT-MUSE obtains the best complete-task average. The gain is largest when a policy must resolve local geometry or maintain alignment under ambiguous vision like LB, IHo and IHD. VT-MUSE is not uniformly best: ViTaL performs better on PoK. This task-level variation suggests that temporal reconstruction is most useful when success depends on sustained, interpretable contact rather than on a short and visually clear transition.

We also ran an exploratory physical-robot evaluation with a Flexiv Rizon 4s and XenseGripper. VT-MUSE performed significantly on all tasks. The four tasks stress different feedback modes. Tube insertion requires correcting pose from intermittent bilateral contact; wiping requires maintaining contact while moving tangentially across the surface; pressing toaster requires continuous force and instant reaction; drawer pull-out needs constant motion direction. Demonstrations are collected with force-feedback teleoperation. This setup tests whether the pretrained temporal abstraction is usable under a modest deployment budget, although it does not isolate sim-to-real transfer from additional real-world pretraining. And the scenarios we pick match the daily life reality.

C. Ablation

Table II(a) varies the sensory window while retaining the rest of the training recipe. Performance peaks at five sampled observations and is not monotonic: increasing the window to six reduces average SR to 41.25%. A longer history therefore does not automatically provide a better state; it can add irrelevant phases of the interaction and make optimization harder. We use $L = 5$ in all main experiments. This result motivates adaptive sampling or learned memory selection rather than simply extending a fixed context.

The compact ablation in Table II(b) keeps the policy and data fixed. Removing either Stage II reconstruction target reduces average SR to 38.75% or 37.00%, showing that global appearance and local contact changes provide complementary supervision. Using Stage I alone performs only slightly above vision-only ACT, while removing temporal contrast lowers the full model by 13 points. Together, these results support the central design choice: align the modalities first, then learn a tactile-conditioned temporal latent through multimodal prediction. The ablation is stated at the objective level; it does not remove the temporal Transformer itself. The pattern also clarifies what the auxiliary losses do not show. Reconstruction metrics alone need not predict control quality: an image decoder can favor pixel fidelity without preserving the contact variables needed for action. We therefore use rollout SR as the primary ablation measure and treat the decoders as training-time probes. A useful next diagnostic would evaluate latent sensitivity to controlled occlusion, slip, and pose perturbations independently of the policy.

D. Discussion and Limitations

The both simulation and real world results and objective ablations are consistent with the intended mechanism: the representation helps most when tactile evolution disambiguates a partially observed geometric state. The weaker PoK result suggests a boundary condition: when the useful event is brief and the subsequent motion is visually simple, a learned reconstruction memory can add less value than a direct aligned encoder. Evaluating event-centered windows would test this interpretation.

The current results do not establish broad generalization. Pretraining and evaluation share a sensor configuration and a small task family. The discrete task token also assumes that task identity is known. Future evaluations should test unseen

TABLE I

MANIPULATION RESULTS. (A) SUCCESS RATE (%) BY TASK; A DASH DENOTES AN UNREPORTED RESULT. (B) AVERAGE SUCCESS RATE FOR REPRESENTATION ABLATIONS.

(a) Simulation setting

Method	LB	PoK	IHo	IHD	Avg.	w/o IHD
ACT (Pure Vision)	42	28	19	15	26	30
ACT+UniVTAC	71	45	23	17	39	46.33
ViTaL*	72	**47**	25	6	37.5	48
FTP-$\pi_{0.5}$*	77	30	47	-	-	51.33
VT-MUSE (Ours)	**84**	38	**68**	**31**	**55.25**	**62.33**

(b) Real world setting

Method	IT	WB	PoD	PT	Avg.
ACT	1/20	3/20	13/20	4/20	26.25
ACT (V+T)	1/20	5/20	15/20	4/20	31.25
VT-MUSE (Ours)	**19/20**	**19/20**	**20/20**	**18/20**	**95**

TABLE II

ABLATION RESULTS. (A) SUCCESS RATE (%) AFFECTED BY CONTEXT WINDOW LENGTH. (B) SUCCESS RATE FOR REPRESENTATION ABLATIONS.

(a) Window length ablation

Window	LB	PoK	IHo	IHD	Avg.
3	60	16	56	14	36.50
4	60	23	57	20	40.00
5	**84**	**38**	**68**	**31**	**55.25**
6	68	18	61	18	41.25

(b) Training objective ablation

Variant	Avg. SR (%)
w.o. tac.pred.	38.75
w.o. vis.recon.	37.00
Stage I only	27.25
w.o. temporal	42.25
VT-MUSE	**55.25**

objects, sensor changes, and task-agnostic inference, with uncertainty estimates and more physical trials. Finally, the fixed-rate temporal window is a practical simplification; the sensitivity result indicates that selecting informative contact events may be more robust than retaining every recent observation. So a good further direction is that establish broad tasks suite both in simulation and real world for greater scale pretraining and fit a VLA or WAM framework for better generalization.

V. CONCLUSION

VT-MUSE learns a deployable visuotactile state by aligning synchronized histories and reconstructing masked recent vision together with tactile depth flow. The representation improves a lightweight action-chunking policy on four simulated tasks and four real world tasks, which shows great effect in different suite domains. While we still claim the need for adaptive memory and broader cross-task pretraining to find more scalable results.

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